HEART DISEASE ANALYSIS

Final Project Report

Dataset: Heart_new2.csv

Tools: Tableau Desktop, Tableau Public, Python, Flask, Pandas, HTML, CSS, JavaScript

Academic Year: 2026–2027

1. INTRODUCTION

Project Overview

The Heart Disease Analysis project is developed to analyze patient healthcare records using the Heart_new2 dataset. Interactive Tableau dashboards and stories are integrated with a Flask web application to provide healthcare insights.

Purpose

The purpose of the project is to help healthcare professionals identify heart disease risk factors, analyze patient demographics, and support evidence-based clinical decisions.

2. IDEATION PHASE

Problem Statement: Healthcare organizations often lack efficient analytical tools to identify heart disease risk factors.

Empathy Map: Target users include cardiologists, healthcare analysts, and hospital administrators.

Brainstorming: Ideas included data preprocessing, KPI dashboards, age, BMI, smoking, diabetes, and physical activity analysis.

3. REQUIREMENT ANALYSIS

Customer Journey Map: Users access the dashboard, review KPIs, analyze risk factors, explore Tableau Story, and make informed decisions.

Solution Requirements: Dashboard visualization, KPI cards, interactive filters, reporting, security, reliability, scalability.

Data Flow: Heart_new2 Dataset → Data Cleaning → Tableau Dashboard → Tableau Story → Flask Application → Users.

Technology Stack: Python, Pandas, Tableau, Flask, HTML, CSS, JavaScript, GitHub.

4. PROJECT DESIGN

Problem–Solution Fit: The proposed solution converts healthcare data into interactive dashboards.

Proposed Solution: Tableau dashboards and stories embedded in Flask.

Solution Architecture: Dataset → Preprocessing → Dashboard → Story → Web Application → Healthcare Professionals.

5. PROJECT PLANNING & SCHEDULING

Four sprint-based development:

Sprint 1 - Data Collection & Preprocessing

Sprint 2 - Dashboard Development

Sprint 3 - Insights & Story

Sprint 4 - Deployment, Testing & Documentation.

6. FUNCTIONAL AND PERFORMANCE TESTING

Verified data rendering, preprocessing, calculated fields, dashboard filters, Tableau Story functionality, and KPI calculations. Performance testing confirmed successful dashboard operation.

7. RESULTS

The project provides KPI dashboards, age category analysis, BMI analysis, smoking analysis, diabetes analysis, physical activity analysis, and general health analysis. Interactive dashboards enable healthcare professionals to identify disease trends and high-risk patients.

8. ADVANTAGES & DISADVANTAGES

Advantages:

• Interactive dashboards
• Evidence-based healthcare insights
• Easy visualization
• Faster decision making

Disadvantages:

• Depends on dataset quality
• No real-time hospital integration
• Predictive analytics not implemented.

9. CONCLUSION

The Heart Disease Analysis project successfully transforms the Heart_new2 dataset into meaningful visual analytics using Tableau and Flask, supporting efficient healthcare decision-making.

10. FUTURE SCOPE

Future work includes machine learning prediction, AI-assisted diagnosis, EHR integration, cloud deployment, mobile applications, and real-time healthcare analytics.

11. APPENDIX

Source Code

Dataset: Heart_new2.csv

GitHub Repository

Project Demonstration Link

Tableau Dashboard Screenshots

Tableau Story Screenshots